Duc Le

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First-year mechanical engineering student at Rutgers with hands-on experience in CAD, FEA, and competition robotics.

EDUCATION

Rutgers University–New Brunswick Expected May 2030

B.S. Mechanical Engineering GPA: N/A

Relevant coursework:

ATLANTIC COUNTY INSTITUTE OF TECHNOLOGY June 2026

Academy of Math, Engineering, and Science GPA: 4.2

PROJECTS

FEA & GD&T | Personal Project October 2026 - Present

- Applying finite element analysis in Onshape to check that 3D-printed robot parts survive impact loading, using the VEX robot's aligner as the case study
- Converting measured robot mass and speed into an estimated impact load, then comparing simulation stress against a hand calculation
- Documenting fits and tolerances on the aligner with a dimensioned drawing and GD&T callouts for printed parts

Volumetric Display | Collaborative Project August 2026 - Present

- Collaborating on a swept-surface 3D display where a fast-spinning surface and synced LEDs form an image viewable from any angle; handling mechanical design in Onshape
- Redesigned the rotating surface from a solid wall to 20 lightweight radial arms to cut drag; cut one wall segment's print time from ~10 hours to 1–2 by hollowing it into a lattice
- Ran FEA on the center rod at ~4,500°/s and designed 3-point snap-fit joints to spread load across the arms

Quadruped Robot | Personal Project October 2026 - Present

- Collaborating on a four-legged robot modeled on the Boston Dynamics dog, built to learn how legged robots walk and balance using real-time tracking

Sprocket | Team Leader & Lead Designer Sept 2025 – Apr 2026

- Led CAD design in Onshape and AutoCAD for a 4-person team, building a competition robot from scratch that solved prior teams' failures in weight, navigation drift, and documentation
- Designed the sensor mounts and tracking-wheel hardware for a closed-loop odometry and PID navigation system
- Won Judge's Award for engineering process and documentation; placed 3rd at SkillsUSA

Launch Vehicle Project | Lead Designer, EGR 101 (Embry-Riddle) Sept 2024 – May 2025

- Led design of a 125 ft two-stage orbital rocket concept for a 4-person team, sized to carry a 4,500 lb payload to low Earth orbit
- Ran structural calculations for max dynamic pressure, sizing a 1/16-inch Aluminum 6063-T5 skin and titanium matrix composite bulkheads, and built a spreadsheet to check applied stress against allowable
- Modeled staging and orbital trajectory in MATLAB and the vehicle in SolidWorks

TECHNICAL SKILLS

CAD & Analysis: Onshape (FEA), SolidWorks, AutoCAD

Design & Fabrication: Mechanical Design, 3D Printing (FDM), Reverse Engineering

Programming & Tools: C++, Python, MATLAB, Excel / Google Sheets

Languages: English, Vietnamese, Chinese

HONORS & AWARDS

Top 10 Academic Student: Recognized for academic excellence at ACIT.